

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and evaluated **AI platform architecture patterns** for enterprise system integration, collaborating with senior architects to assess **LLMs, RAG systems, agentic frameworks, and guardrails** against Marsh's existing technology landscape; produced **reference architecture diagrams and technology radar documents** that guided adoption decisions across 8+ business units.
- Architected **system design for AI-driven operational dashboards** using **Power BI, DAX, and data modelling** to consolidate enterprise metrics; integrated **AWS cloud platforms (EC2, Lambda, CloudWatch)** with **REST API integrations** across interconnected systems, reducing reporting latency by **60%** and enabling real-time visibility for stakeholders.
- Conducted hands-on **proofs-of-concept** for emerging AI technologies, partnering with software engineers to prototype **LLM integration patterns, embeddings, and memory systems**; documented architecture decisions in formats consumable by both human engineers and **AI coding agents (GitHub Copilot, Claude Code, Cursor)**, establishing reusable templates for **8 subsequent PoCs**.
- Analyzed enterprise architecture across **ERP systems, business-system integrations, and change-control workflows** to inform AI adoption strategy; mapped **system dependencies, data flows, and regulatory constraints (GDPR, audit-trail requirements)** across legacy and cloud infrastructure, enabling architects to recommend AI solutions that aligned with existing governance.
- Contributed to **architecture documentation and presentation frameworks** for Marsh's Architecture Council and Dev Council forums; authored **high-level integration diagrams, decision logs, and technology recommendations** covering **cloud platform selection (AWS, Azure, GCP), AI/ML capabilities, and enterprise integration patterns**; presented findings to **3 cross-functional stakeholder groups**.
- Demonstrated strong **intellectual curiosity and rapid technology evaluation** by actively tracking emerging AI/ML innovations (**LLMs, vector databases, orchestration frameworks**); built and tested prototypes using modern AI-first languages and platforms, synthesizing technical findings into architecture recommendations that balanced innovation speed with enterprise risk management.

SKILLS

Architecture & Design – system architecture, AI platform design, reference architectures, architecture documentation, enterprise architecture, technology radar, integration patterns, design thinking.

AI/ML Technologies – LLMs, large language models, RAG (retrieval-augmented generation), embeddings, agentic frameworks, guardrails, vector databases, prompt engineering, AI coding platforms.

AI Coding & Development Tools – GitHub Copilot, Claude Code, Cursor, Python, Java, Spring Boot, REST APIs, FastAPI, Flask.

Cloud Platforms – AWS (EC2, S3, Lambda, CloudWatch), Azure, GCP, cloud infrastructure design, cloud integration patterns.

Data & Analytics – Power BI, Tableau, DAX, Power Query, data modelling, data warehousing, ETL, SQL (PostgreSQL, MySQL, Oracle), pandas, NumPy, scikit-learn.

Enterprise Systems & Integration – ERP systems, enterprise business systems, system integrations, change-control management, GDPR, audit-trail requirements, regulatory compliance.

DevOps & Infrastructure – Docker, Linux, Git, Jenkins, CI/CD pipelines, Splunk, Dynatrace, CloudWatch, monitoring and observability.

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin	02/2026
BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India	01/2021

PROJECTS

AI CoE Reference Architecture & Technology Radar Platform (HCL Technologies)

- Designed and documented a **multi-layered AI platform reference architecture** covering **LLM integration, RAG systems, agentic orchestration, and guardrail deployment patterns** across Marsh's **AWS and Azure cloud estates**; created **architecture decision records (ADRs) and system design diagrams** in formats optimized for both human architects and **AI coding agents**.
- Built a **technology evaluation framework** for assessing emerging AI/ML technologies; conducted **12 hands-on proof-of-concept sprints** testing **LLM providers, vector databases, orchestration frameworks, and memory systems**; documented findings in a **dynamic technology radar** consumed by **3 architecture forums and 50+ engineers**.
- Architected **enterprise integration patterns** that mapped **AI capabilities to existing ERP systems, business-system data flows, and regulatory constraints (GDPR, audit-trail requirements)**; reduced architecture review cycles by **40%** through reusable integration templates and cross-system dependency documentation.
- Established **architecture documentation standards** for AI adoption, creating templates and guidelines that enabled both human review and **AI agent comprehension**; enabled **8 project teams** to independently architect AI solutions while maintaining enterprise consistency.

Enterprise Operational Intelligence Dashboard – Banking Payments & Regulatory Compliance (HCL Technologies)

- Designed and deployed a **Power BI enterprise dashboard architecture** integrating **SWIFT messaging operations (MT/MX formats), ISO 20022 transaction flows, and payments-processing SLAs** across PostgreSQL, Oracle, and MySQL data sources; dashboard consumed by **150+ banking operations staff and regulatory compliance teams**.
- Architected **ETL and data modelling strategy** using **Python, SQL, and Power Query** to consolidate transaction-level payment data, regulatory audit trails, and GDPR compliance metadata; reduced regulatory reporting time by **65%** and enabled real-time visibility into **cross-border SEPA and instant-payment flows**.
- Integrated **AWS CloudWatch and Splunk monitoring** with **DAX measures and dynamic Power BI visuals** to surface P1/P2 incident patterns, SLA breaches, and payment-message routing anomalies; enabled operations to resolve **payment-flow issues 3x faster** through data-driven root-cause analysis.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)